Task-14:- Technical Documentation (VPC networking setup)

1. Introduction

This document describes the design and implementation of a **multi-VPC network architecture in AWS** for ABC Technologies. The objective is to simulate a company network with one Headquarters (HQ) and four branch offices connected using **VPC Peering across regions**.

The HQ network is deployed in **US East (N. Virginia)** while branch offices are deployed in **US East (Ohio)**.

This setup demonstrates how organizations can design **inter-office connectivity in the cloud** using multiple VPCs.

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2. Objective

Design and implement a multi-VPC AWS network environment across two regions and establish connectivity using **Amazon VPC Peering**.

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3. Problem Statement

ABC Technologies is expanding its infrastructure and needs a network architecture where:

- HQ office is hosted in **us-east-1**
- Branch offices (A, B, C, D) are hosted in **us-east-2**
- Each office has a **separate VPC**
- All branch offices communicate with HQ using **VPC Peering**

This architecture helps the company understand **multi-VPC networking and cross-region communication**.

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4. Network Design

HQ Office

Region: **us-east-1**
IP Type: **Class A Private Network**

VPC CIDR:

```
10.0.0.0/16
```

Subnets:

Subnet	CIDR	Hosts
Server Subnet	10.0.0.0/24	256
Management Subnet	10.0.1.0/24	256

Both subnets support the required **200 hosts**.

Branch Offices

Region: **us-east-2**

Branch offices use **Class B private IP ranges**.

Office	CIDR	Hosts
Office A	172.16.0.0/24	256
Office B	172.16.1.0/24	256
Office C	172.16.2.0/24	256

Office D	172.16.3.0/24	256

Each subnet supports more than **50 hosts**.

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5. AWS VPC Environment Setup

Step 1: Create HQ VPC

Region:

```
us-east-1
```

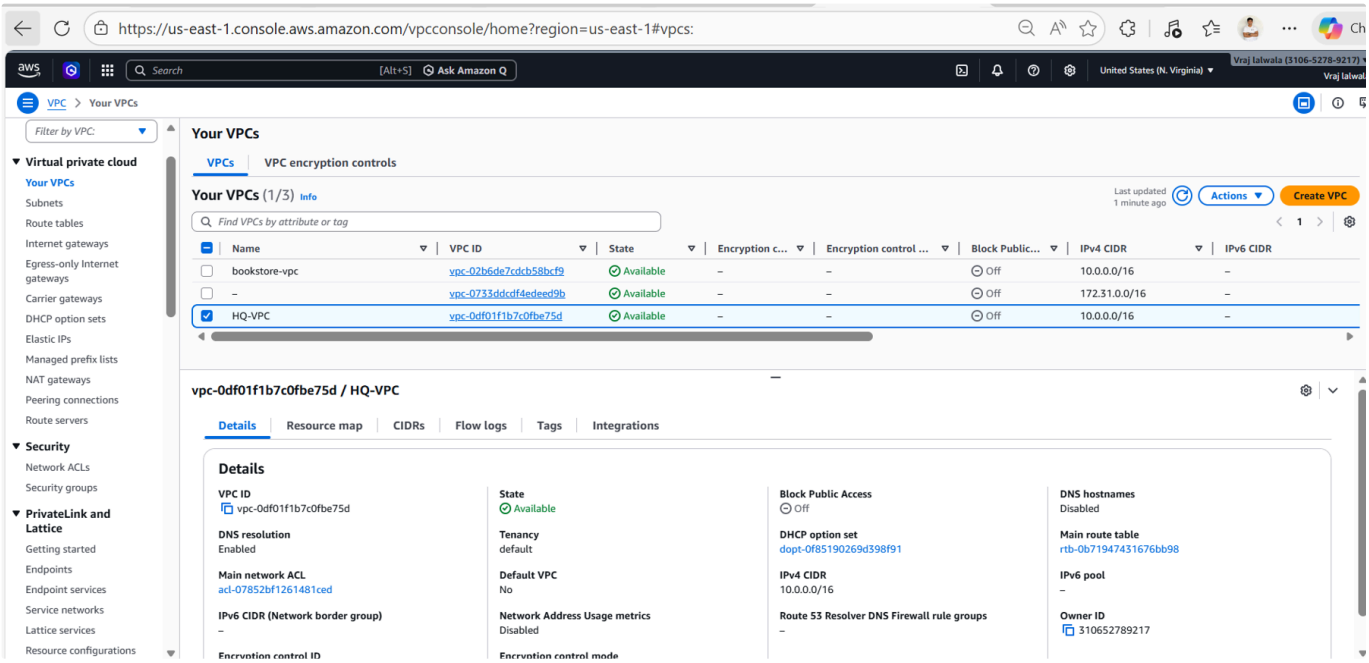
CIDR:

```
10.0.0.0/16
```

Subnets created:

Subnet	CIDR
HQ Server Subnet	10.0.0.0/24
HQ Management Subnet	10.0.1.0/24

Screenshot Placement:-



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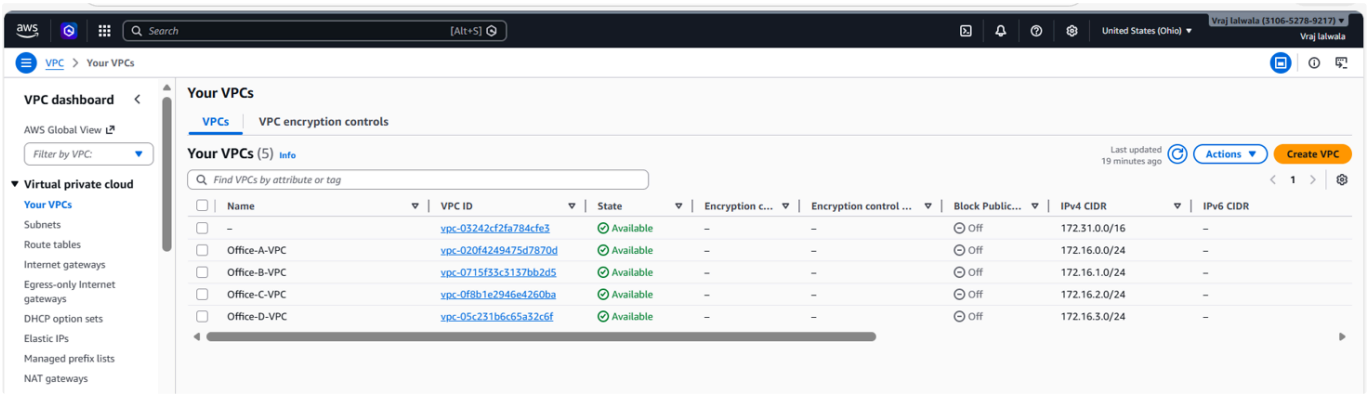
6. Branch Office VPC Setup

Branch VPCs were created in **us-east-2**.

Office	VPC CIDR
Office A	172.16.0.0/24
Office B	172.16.1.0/24
Office C	172.16.2.0/24
Office D	172.16.3.0/24

Each VPC contains one subnet.



7. Subnet Configuration

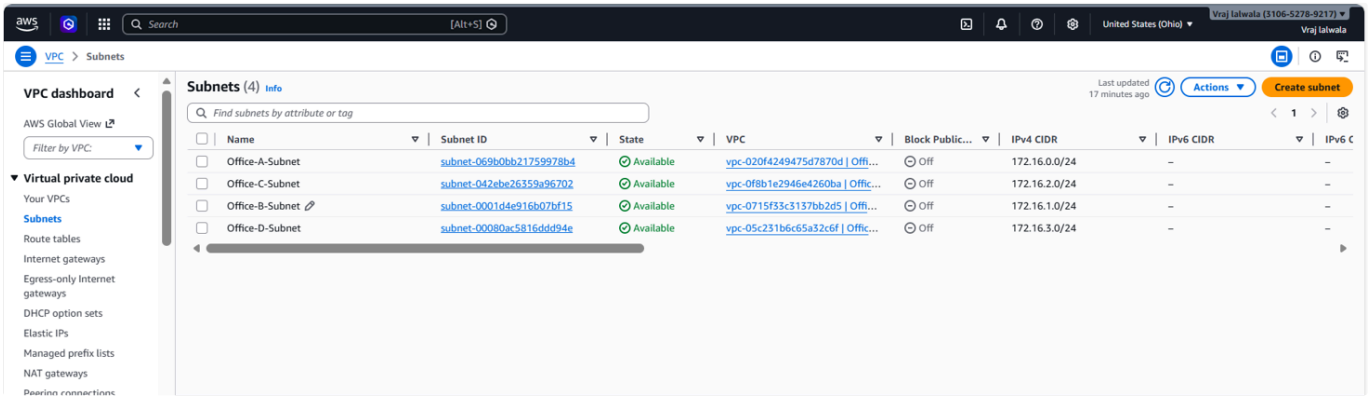
Subnets were created according to the host requirements.

HQ Subnets

Subnet	CIDR
HQ Server	10.0.0.0/24
HQ Management	10.0.1.0/24

Branch Subnets

Office	CIDR
Office A	172.16.0.0/24
Office B	172.16.1.0/24
Office C	172.16.2.0/24
Office D	172.16.3.0/24

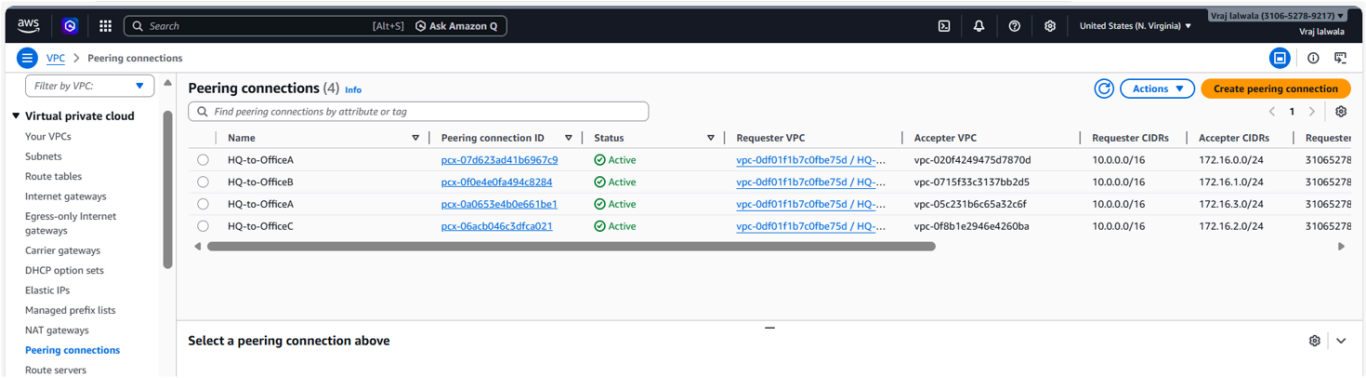


8. VPC Peering Configuration

Cross-region VPC peering connections were created between HQ and each office.

Peering Name	Connection

HQ-to-OfficeA	HQ ↔ Office A
HQ-to-OfficeB	HQ ↔ Office B
HQ-to-OfficeC	HQ ↔ Office C
HQ-to-OfficeD	HQ ↔ Office D



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9. Route Table Configuration

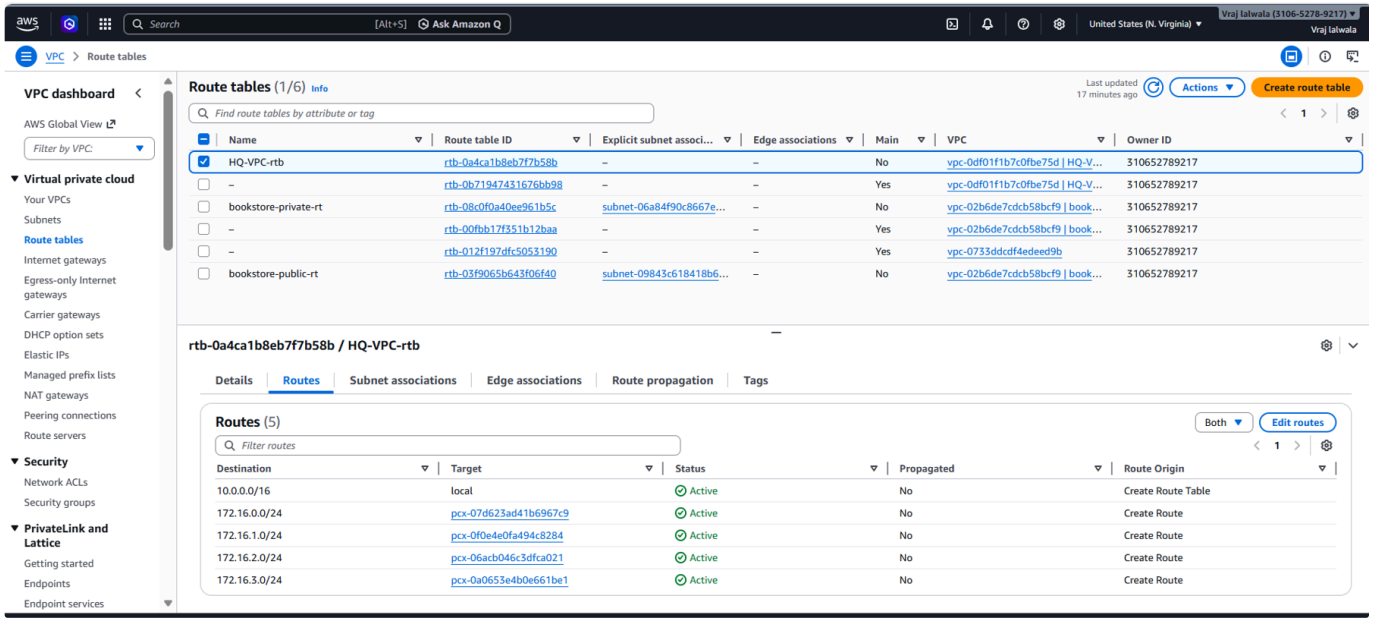
Routes were updated so that each VPC can communicate with HQ.

Example HQ Route Table:

Destination	Target
172.16.0.0/24	Peering connection
172.16.1.0/24	Peering connection
172.16.2.0/24	Peering connection
172.16.3.0/24	Peering connection

Branch route tables contain routes pointing to:

```
10.0.0.0/16 → VPC Peering
```



10. EC2 Deployment

EC2 instances were launched in:

Location	Subnet
HQ Server	10.0.0.0/24
HQ Management	10.0.1.0/24
Office A	172.16.0.0/24
Office B	172.16.1.0/24
Office C	172.16.2.0/24
Office D	172.16.3.0/24

Instances were used to verify network connectivity.

11. Security Group Configuration

Security groups were configured to allow:

```
SSH (Port 22)
ICMP (Ping)
```

Source:

```
All VPC CIDR ranges
```

This allows testing between offices.

12. Connectivity Testing

Connectivity between HQ and branch offices was tested using:

```
telnet <private-ip> 22
```

Example:

```
telnet 172.16.0.10 22
```

Successful response confirms that **VPC Peering routing works correctly**.

Screenshot Placement

📸 Insert screenshot showing **connectivity test result**

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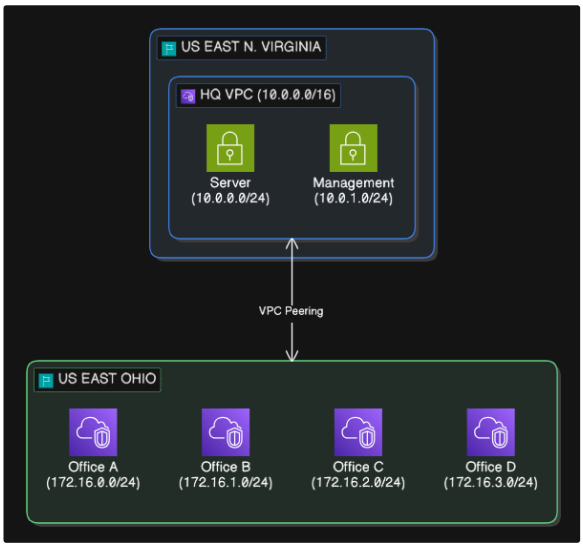
13. Network Architecture Diagram




```
                HQ VPC (us-east-1)
                    10.0.0.0/16
                   /           \
          Server Subnet      Mgmt Subnet
            10.0.0.0/24      10.0.1.0/24
                |
                |
        -------------------------
        |       |        |       |
  OfficeA OfficeB OfficeC OfficeD
172.16.0 172.16.1 172.16.2 172.16.3
   VPC      VPC      VPC      VPC
```

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14. Conclusion

This project successfully demonstrated:

- Multi-VPC architecture in AWS
- Cross-region VPC connectivity
- Route table configuration
- Network segmentation
- Inter-office communication testing

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15. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____